In-Car Voice Assistant

Ablation and robustness report

Target speaker extraction -> presence -> wake word -> intent

Study date	Evidence status	Training
2026-10-09	**Proxy diagnostics**	**Not started**

Executive finding

The ablations show a clear safety-versus-recall tradeoff. The presence gate is the main control on false actions: lowering it recovers target wakes but admits passenger/background events. At the current frame-presence operating point, the full candidate path is conservative (25% proxy wake recall, 12.5% proxy intent accuracy, and 0 false actions/hour). The best proxy intent operating point found in the threshold sweep reached 37.5% intent accuracy, but at 128.6 false actions/hour. These are not real-driver reliability scores because the workspace still has zero continuous same-driver recordings.

0	8	120 s	12 passed
real-driver sessions	proxy scenes per run	AISHELL target-absent check	tests (1 skipped)

Important evidence boundary

The positive scenes use a Marvin clip and an FSC command clip from different public corpora. Their speaker identity is not verified. AISHELL-5 supplies a real in-car target-absent diagnostic only. No result in this PDF should be described as a real-driver end-to-end accuracy or production false-accept rate.

1. Study design and data status

Question

Can the assistant keep a target driver through enrollment, detect Marvin, understand the following command, and avoid acting on passenger or background speech? The requested real-driver study requires the same speaker's enrollment, wake word, and command in continuous cabin recordings, plus passenger and no-command intervals.

Evidence source	Used for	Status / limitation
Real custom in-car sessions	Positive driver evaluation	0 sessions; collection still required
Cross-corpus proxy	Continuous cascade debugging	8 scenes per run; speaker consistency false
AISHELL-5 Eval2	Real-cabin target-absent rejection	120 seconds; no positive wake/command
CAVEMOVE	Cabin noise and multichannel mixtures	Augmentation material, not driver sessions

The official AISHELL-5 release identifies the dataset as CC BY-SA 4.0: https://us.openslr.org/159/. Commercial or sign-in datasets were not treated as evidence without project permission.

Proxy scene construction

- 14 seconds per scene, with target wake at about 1 second and target command at about 2.1 seconds.
- Passenger Marvin near 6 seconds and passenger non-wake speech near 9 seconds.
- CAVEMOVE multichannel cabin noise at -5, 0, 10, and 20 dB cabin SNR.
- Passenger loudness sweep at -10, 0, and +10 dB relative to the target.
- Stage 1 context is 3 seconds and stream hop is 1 second; reported wake latency is window-center latency, not measured wall-clock response time.

Current candidate checkpoints

Component	Checkpoint
spex	stage1_spex_transfer_mc_cavemove_cabin/best.pt
presence	stage2_frame_presence_gsc_aishell_ft1/best.pt
kws	stage3_kws_gsc_marvin_tse_ft2/best.pt
target_kws	stage3_target_kws_gsc_marvin_ft1/best.pt
intent	stage4_intent_wav2vec_ft8_resume_tse_domain/best.pt

2. Component ablation matrix

This is the primary same-scene ablation at passenger SNR 0 dB using the target-KWS candidate configuration. Each row uses identical scenes and differs only in the named component. False actions are post-wake command events outside the labeled target command region; they are a proxy diagnostic, not a human action-rate estimate.

Variant	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Full	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No TSE	50.0%	50.0%	64.3/h	1.75 / 2.35 s
No presence	100.0%	12.5%	417.9/h	1.50 / 1.50 s
No speaker matching	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No wake smoothing	100.0%	0.0%	0.0/h	1.25 / 1.50 s
Direct command oracle	-	12.5%	-	- / - s

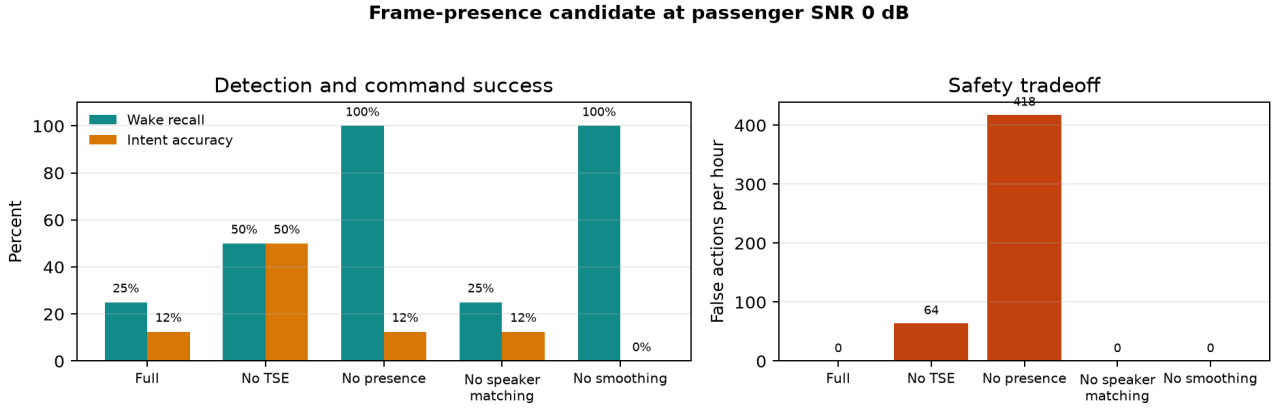


Figure 1. The current frame-presence candidate is conservative. Disabling presence greatly increases recall but produces many proxy false actions.

Interpretation

• Presence checking is safety-critical in this proxy. Removing it produced 100% target wake recall but 417.9 false actions/hour.

• One-hit wake detection produced 100% wake recall but 0% intent accuracy in this run, indicating that accepting a wake is not enough; the post-wake command window is still unstable.

• Speaker matching did not change the aggregate result at this operating point. That result is inconclusive without same-speaker passenger recordings.

• The no-TSE row is an intentionally harsh diagnostic. Its raw mixture is outside the training distribution of downstream heads, so it is not a candidate architecture.

• The direct command row uses labeled command segmentation and is an oracle baseline, not a deployable streaming path.

3. Passenger loudness ablation

The passenger loudness sweep tests whether the safety behavior changes when passenger speech is louder or quieter than the target. Negative SNR means the passenger is louder.

Passenger SNR	Variant	Wake recall	Intent accuracy	False actions/h
-10 dB	Full	25.0%	12.5%	96.4/h
-10 dB	No TSE	50.0%	50.0%	160.7/h
-10 dB	No presence	100.0%	12.5%	450.0/h
-10 dB	No smoothing	100.0%	0.0%	192.9/h
+0 dB	Full	25.0%	12.5%	0.0/h
+0 dB	No TSE	50.0%	50.0%	64.3/h
+0 dB	No presence	100.0%	12.5%	417.9/h
+0 dB	No smoothing	100.0%	0.0%	0.0/h
+10 dB	Full	25.0%	12.5%	0.0/h
+10 dB	No TSE	50.0%	50.0%	32.1/h
+10 dB	No presence	100.0%	12.5%	289.3/h
+10 dB	No smoothing	100.0%	0.0%	0.0/h

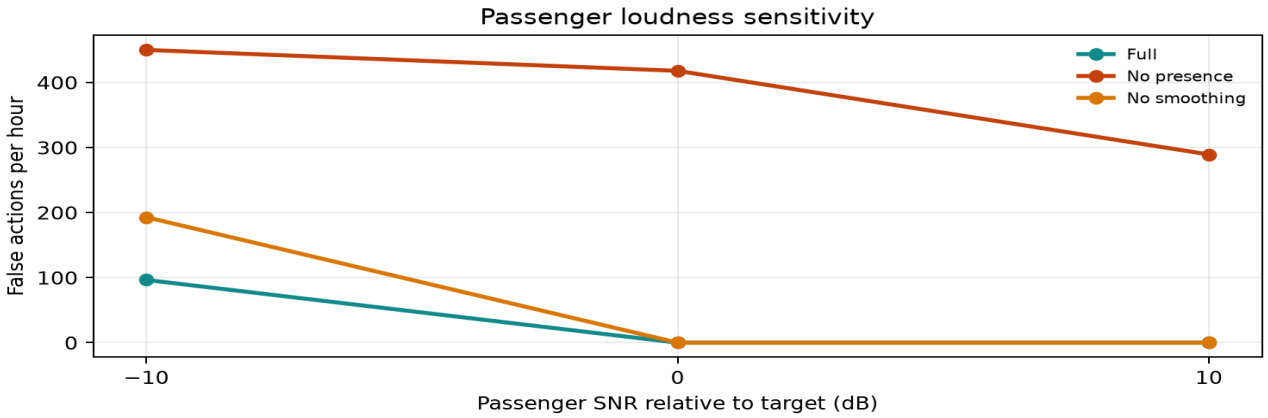


Figure 2. Louder passenger speech increases proxy false actions, especially when the presence gate is removed.

4. Threshold operating-point ablation

The grid replayed presence thresholds 0.30, 0.50, 0.70, and 0.90 against KWS thresholds 0.30, 0.40, 0.50, and 0.60. The expensive Stage 1 scores were computed once per scene and the state machine was replayed for each cell.

Presence	KWS	Wake recall	Intent accuracy	False actions/h
0.30	0.50	100.0%	37.5%	321.4/h
0.50	0.50	100.0%	37.5%	128.6/h
0.70	0.50	25.0%	12.5%	0.0/h
0.90	0.50	0.0%	0.0%	0.0/h

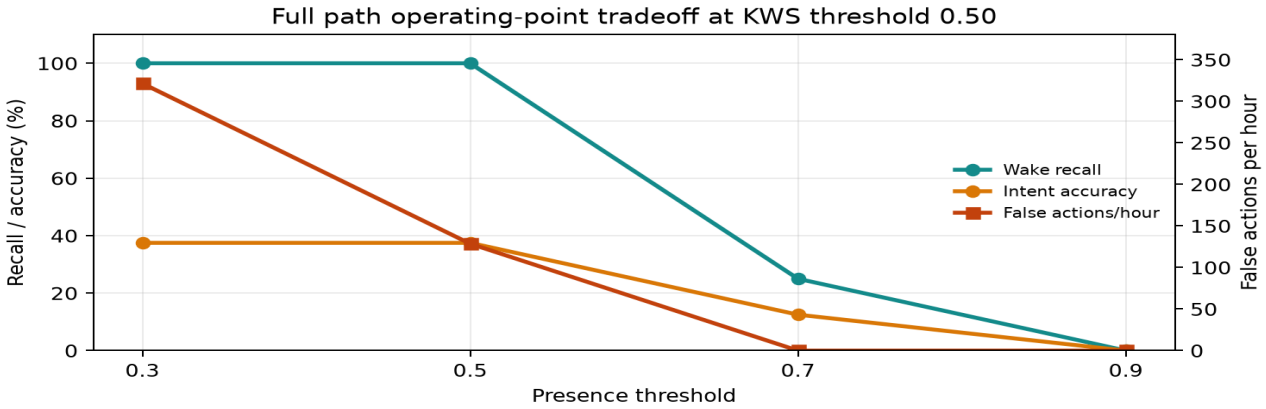


Figure 3. Lowering the presence threshold improves proxy recall, but false actions rise sharply. KWS threshold changes were less influential than presence threshold changes in this grid.

Practical reading

The current p=0.70 operating point is a conservative starting point for safety, not a claim of good recognition. The p=0.50 result is a candidate only for tuning experiments because its false-action rate is unacceptable as a final operating point. Proper threshold selection must use a consented tuning split and then be frozen before held-out testing.

5. Scalar versus framewise presence

The target-conditioned KWS setup was kept fixed while changing the presence head from the candidate framewise checkpoint to the scalar cabin-trained checkpoint.

Presence head	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Framewise	25.0%	12.5%	0.0/h	1.50 / 1.50 s
Scalar	100.0%	25.0%	32.1/h	1.75 / 2.80 s

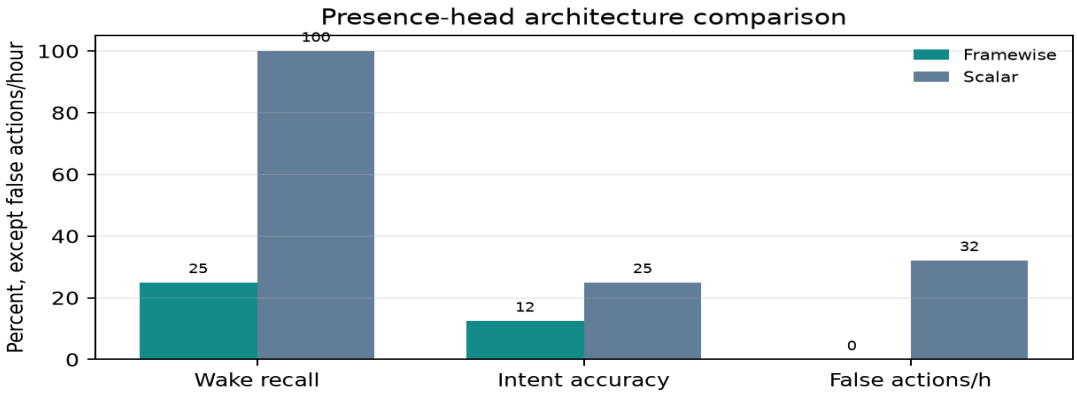


Figure 4. Scalar presence improves proxy recall but admits more false actions and has more variable timing.

This is a useful candidate for real-driver tuning, but not a replacement decision. The proxy does not contain the required continuous same-driver enrollment, passenger, and command events.

6. Real-cabin target-absent diagnostic

AISHELL-5 Eval2 session 001 was evaluated for 120 seconds with a fixed LibriSpeech enrollment. The enrolled target is absent, so this tests rejection only.

Metric	Result
Duration	120 s (0.0333 h)
Speech occupancy	80.6%
Full-path smoothed false events	0.0/h
KWS-only smoothed false events	0.0/h

This result is encouraging for one target-absent real-cabin diagnostic, but it cannot establish driver wake recall, command accuracy, or speaker rejection of a real passenger saying Marvin.

7. Overall assessment and next step

• The TSE/presence cascade is not yet reliable as a single end-to-end system on the current proxy. Its conservative settings suppress many events, including target wakes.

• The immediate engineering problem is calibration and post-wake command capture: lower thresholds recover wakes but create false actions; more smoothing increases latency and can lose the command relationship.

• The scalar presence head is a promising recall-oriented candidate, but it must be tuned on consented driver/passenger sessions before adoption.

• The next highest-value experiment is a small real-driver pilot with speaker-disjoint tuning and test sessions. Without that dataset, further proxy optimization risks overfitting to cross-corpus artifacts.

No model training was started for this report. The report documents inference-only ablations and threshold replays.

8. Reproducibility

```
# From the repository root
python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_2026-10-09.json

python scripts/eval/eval_threshold_ablation_sweep.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --presence-thresholds 0.30 0.50 0.70 0.90 `
  --kws-thresholds 0.30 0.40 0.50 0.60 `
  --variants full no_speaker_matching no_wake_smoothing `
  --output logs/threshold_ablation_sweep_2026-10-09.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_scalar_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json
```

Primary source files

Sources: reports/pilot_evaluation_status_2026-10-09.md; reports/additional_ablation_results_2026-10-09.md; logs/continuous_ablation_matrix_2026-10-09.json; logs/continuous_ablation_matrix_passenger_m10_2026-10-09.json; logs/continuous_ablation_matrix_passenger_p10_2026-10-09.json; logs/threshold_ablation_sweep_2026-10-09.json; logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json; logs/aishell5_target_absent_study_2026-10-09.json